

Contestant Number: \_\_\_\_\_

Time: \_\_\_\_\_

Rank: \_\_\_\_\_



# ADVANCED SPREADSHEET APPLICATIONS (235)

## STATE 2021

### Production:

Job 1: Chart \_\_\_\_\_ (100 points)

Job 2: Range, Table and Sparklines \_\_\_\_\_ (90 points)

Job 3: Pivot Table \_\_\_\_\_ (75 points)

TOTAL POINTS \_\_\_\_\_ (265 points)

### Test Time: 90 minutes

## **GENERAL GUIDELINES:**

*Failure to adhere to any of the following rules will result in disqualification:*

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

## **INSTRUCTIONS:**

1. Key your Member ID and printout number in the right section of a footer on each printout on a separate line in the footer. Your name or initials should not appear on any work you submit.
2. Center the following information in a header on each printout: File name and the sheet name. Each item should appear on a separate line in the header.
3. If you finish before the end of the testing time, notify the proctor. Time may be a factor in determining the winner in the event of a tie.
4. When turning in your contest, arrange all jobs in the correct printout order (included in instructions).
5. Write your Member ID in the provided spaces on the test cover page.

| <b>Job 1 (Chart)</b>                                     | <b>Points Possible</b> | <b>Points Earned</b> |
|----------------------------------------------------------|------------------------|----------------------|
| Contestant # and Job # in right section of footer        | 10                     |                      |
| File name and sheet name in the header                   | 10                     |                      |
| Merge and Center the Title over A1:E1                    | 5                      |                      |
| Bold and Center Header Row Information                   | 5                      |                      |
| Create a 2-D Line Graph with Markers comparing treatment | 15                     |                      |
| Switch Data on the X and Y axis                          | 10                     |                      |
| Label the X and Y axis                                   | 10                     |                      |
| Chart Title                                              | 5                      |                      |
| Legend on the Right of the Chart                         | 5                      |                      |
| Chart color is Colorful Color 4                          | 5                      |                      |
| Filter Chart to show only Control, Heat and Water Bath   | 15                     |                      |
| Data fits to one page                                    | 5                      |                      |
| <b>Subtotal</b>                                          | <b>100</b>             |                      |
| <b>Job 2 (Range, Table and Sparklines)</b>               | <b>Points Possible</b> | <b>Points Earned</b> |
| Contestant # and Job # in right section of footer        | 10                     |                      |
| File name and sheet name in the header                   | 10                     |                      |
| Format data as a Table, Table Style Medium 2             | 10                     |                      |
| Remove table Filters                                     | 5                      |                      |
| Show a table Total Row                                   | 5                      |                      |
| Find the Average for each month                          | 10                     |                      |
| SUM Formula for cells B12:B15                            | 10                     |                      |
| Column Sparkline for each employee                       | 10                     |                      |
| High Point in each Sparkline                             | 5                      |                      |
| Sparkline Style Accent 5, Lighter 40%                    | 5                      |                      |
| Landscape Orientation and fit to page                    | 5                      |                      |
| Print Data and formulas                                  | 5                      |                      |
| <b>Subtotal</b>                                          | <b>90</b>              |                      |

| <b>Job 3 (Pivot Table)</b>                                             | <b>Points Possible</b> | <b>Points Earned</b> |
|------------------------------------------------------------------------|------------------------|----------------------|
| Contestant # and Job # in right section of footer                      | 10                     |                      |
| File name and sheet name in the header                                 | 10                     |                      |
| Pivot Table created correctly                                          | 20                     |                      |
| Filter data to show Male cats and dogs                                 | 10                     |                      |
| PivotChart style changed to Style 3, color changed to Colorful Color 4 | 10                     |                      |
| Correct Title                                                          | 5                      |                      |
| Legend removed                                                         | 5                      |                      |
| X Axis labeled, Animal                                                 | 5                      |                      |
| <b>Subtotal</b>                                                        | <b>75</b>              |                      |
| <b>TOTAL POINTS</b>                                                    | <b>265</b>             |                      |

## Job 1 – Chart

Create the following workbook as shown below. Change font to Times New Roman, size 12.

|   | A                                           | B            | C            | D            | E            |
|---|---------------------------------------------|--------------|--------------|--------------|--------------|
| 1 | <b>Pansy Transpiration - Mass over Time</b> |              |              |              |              |
| 2 | <b>Treatment</b>                            | <b>Day 1</b> | <b>Day 2</b> | <b>Day 3</b> | <b>Day 4</b> |
| 3 | Control                                     | 100          | 82.8         | 72.1         | 60           |
| 4 | Wind (fan)                                  | 100          | 61.7         | 55.6         | 34.8         |
| 5 | Light                                       | 100          | 84           | 60.5         | 50.6         |
| 6 | Heat                                        | 100          | 64.8         | 49.4         | 41.2         |
| 7 | Water Bath                                  | 100          | 98.9         | 97.7         | 96           |
| 8 | Bell Jar                                    | 100          | 99.3         | 98.2         | 97.6         |

- Rename Sheet 1, “Pansy Transpiration”.
- Save the Spreadsheet as “Science Data”.
- Merge and Center the Title information in Cells A1:E1 over the data.
- Format the Heading Row to Bold and Center.
- Create a 2-D Line Graph with Markers comparing the Treatment over all 4 days.
- Switch the data on the X and Y axis, so that Day 1 shows 100 for all Treatments.
- Label the X and Y Axis – (Time and Mass)
- The Chart Title should be: “Pansy Transpiration”
- Move the Legend to the Right of the Data.
- Change the chart color to Colorful Color 4.
- Filter the Chart so only Control, Heat and Water Bath are visible.
- Position the Chart below the data.
- Make sure it fits to one page and print.

## Job 2 – Range, Table and Sparklines

- Open the **Sales Interactions** workbook from the folder.
- Rename Sheet 1, “2021 Sales”.
- Select the Range C4:E8 and name the Range, Quarter1.
- Repeat and Name Ranges for the rest of the year, Quarter2, Quarter3 and Quarter4.
- Type the following information into the spreadsheet in cells A10:A13:
  - Quarter 1 Sales, Quarter 2 Sales, Quarter 3 Sales, Quarter 4 Sales
- Resize column A so all of the data fits.
- Select the data in cells A3:N8 and format as a Table, using the Table Style Medium 2. Keep it in the same location.
- Remove the Filters for the table.
- Show a Total Row for the table.
- Find the Average for each month in the Table.
- In cells B12:B15, enter a SUM formula and reference the corresponding Range of cells for each quarter.
- In Cell O3, type “Sparkline”.
- Create a Column Sparkline in Cell O4 for the entire year for Ferne Amos.
- Show a Column Sparkline for each employee in Column O.
- Show the High Point of each Sparkline.
- Change the Sparkline Style to Accent 5, Lighter 40%.
- Change the Orientation to Landscape and fit the data to one page.
- Print the data.
- Show formulas and resize the columns so they shrink to the size of the data.
- Hide Column A and Column O.
- Print formulas.

### Job 3 – Pivot Table

- Open the **Parkplace Animal Clinic** workbook from the folder.
- Rename Sheet 1, Animals.
- Using the data in A2:D54, Create a Pivot Chart. Place it in a New Worksheet.
- Rename the new Sheet, Male Animals.
- Filter by Gender and Breed. The Axis (Categories) should be Type. The Values should find the Count of the Gender.
- Filter the data so it only shows only the Male cats and dogs.
- Move the Pivot Chart below the data.
- Change the Pivot Chart Style to Style 3 and the color to Colorful Color 4.
- Update the title to MALE ANIMALS.
- Remove the legend from the Pivot Chart.
- Add an X axis and label it, Animal.
- Print the Male Animal sheet only.